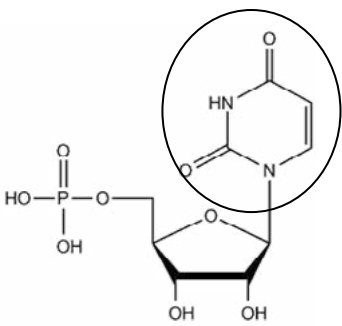


| Question number | Answer   | Mark |
|-----------------|----------|------|
| 2(a)(i)         | D ribose | (1)  |

| Question number | Answer                                                                            | Mark |
|-----------------|-----------------------------------------------------------------------------------|------|
| 2(a)(ii)        |  | (1)  |

| Question number | Answer                                                               | Mark |
|-----------------|----------------------------------------------------------------------|------|
| 2(a)(iii)       | <ul style="list-style-type: none"> <li>• synthesis of RNA</li> </ul> | (1)  |

| Question number | Answer                                                                                                                                                                                                                         | Additional guidance                       | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------|
| 2(b)(i)         | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• OMP decarboxylase acting as a biological catalyst (1)</li> <li>• it lowers the activation energy of this reaction (1)</li> </ul> | Allow forms an enzyme - substrate complex | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                             | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 2(b)(ii)        | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• OMP decarboxylase is specific (for this substrate) (1)</li> <li>• because the active site of this enzyme has a particular shape (1)</li> <li>• therefore binds only with orotidine monophosphate (1)</li> </ul> | (3)  |